

Trust Flow-Driven Trusted Federated Learning in Edge Computing: A Study on Key Technologies

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Abstract. Federated learning in edge environments faces compound risks from data heterogeneity, model poisoning, and runtime trust uncertainty. This paper presents a trust flow-driven trusted federated learning framework that unifies trusted report verification, content auditing, historical reputation evolution, risk-state updating, and layer-wise robust aggregation into a closed-loop pipeline. The proposed method combines hard attestation gating with exponential trust attenuation, introduces a server-prioritized clean reference for contribution validation, and adopts orthogonal dual-stream updates (history stream and risk stream). During aggregation, risk-gated layer-wise admission, survivor re-normalization, and dynamic clipping are jointly applied. We implement the framework on Flower and evaluate it on CIFAR-10 with 20 clients under six mixed attack types. Statistical results over 5 independent runs ($n = 5$) show that malicious clients are effectively identified with negligible permanent impact on benign nodes. The framework achieves an average accuracy of 92.31% ($\pm 0.09\%$) and maintains an average ASR of 10.21% ($\pm 0.05\%$), with a representative benchmark runtime of 3878.53s. The results indicate that trust-flow orthogonal modeling provides robust malicious-client detection while maintaining practical convergence under heterogeneous edge settings.

Keywords: Federated Learning · Edge Computing · Trusted Execution Environment · Risk State Modeling · Layer-wise Robust Aggregation · Backdoor Defense

1 Introduction

As an emerging distributed machine learning paradigm, federated learning (FL) provides an ideal solution for data silos and privacy protection requirements in edge computing scenarios through the mechanism of “data remains local, only model parameters are exchanged” [1]. In edge applications such as the Industrial Internet of Things (IIoT), medical diagnostics, and autonomous driving, rich perceptual data generated by massive smart devices can be used to collaboratively train high-performance global models without exposing the raw data.

However, when federated learning transitions from idealized data centers to realistic edge computing topologies, the system inevitably faces compound

security challenges intertwined with extreme environmental uncertainty, non-independent and identically distributed (Non-IID) data, and advanced behavioral fraud. On the “internal concern” side, naturally distributed edge devices not only have limited computing power, but their independently sensed fragmented scenarios result in highly skewed, long-tailed characteristics in local samples. From the perspective of formal optimization, this severe structural bias causes an extremely large variance in the local gradient updates of honest nodes (i.e., $\text{Var}(\|\nabla\mathcal{L}_k\|) \gg \sigma_{threshold}$), which inherently triggers profound global model spatial drift. On the “external threat” side, edge networks without mandatory physical admission lack a root of trust, leaving the vulnerable client tier highly susceptible to adversary exploitation. Malicious participants can implement targeted, deep-compositional model poisoning and stealthy backdoor attacks (such as clean-label poisoning and deep semantic backdoors) [2,3]. Even more deadly, in severely overlapping macroscopic scenarios, facing the high-perturbation background provided by “internal concerns”, adversaries deliberately construct high-order poisoned gradients that meet stealthy fine-tuning constraints (i.e., making $\|\Delta W_{malicious} - \Delta W_{benign}\| < \epsilon$ for specific backdoor paths), easily bypassing traditional anomaly detection boundaries. The “high-variance honest feature drift” significantly blurs the security identification boundary of conventional distance metric algorithms, causing fundamental defenses that rely on surface parameter statistical distances or single gradients to fall into a performance trade-off dilemma between high false negatives (failing to block advanced latent attacks) and high false positives (mistakenly killing legitimate extreme edge nodes).

Existing active defense mechanisms in federated learning fail to flawlessly resolve the aforementioned dilemmas. The vast majority of studies focus on “Byzantine-Robust” designs at the model aggregation stage, such as Krum, Trimmed Mean, and RFA, which attempt to cull abnormal gradients through distance measurements or geometric median statistical mechanisms [4,5,6]. However, owing to the lack of a root of trust in the underlying physical dimension, these lines of defense are highly prone to failure in highly heterogeneous edge environments. On the other hand, schemes like FLTrust explore cross-round time-dimension trust management [7,8], but when confronted with adversaries adopting advanced “watering hole” strategies of “accumulating fake reputation by lurking early, and continuously poisoning in small batches later”, extensive single-line score management often exhibits critical judgment lags. More fatally, under strong Non-IID environments, the natural gradient direction of honest long-tail nodes inherently forms a large angle with the global reference, leading the core mechanism based on cosine similarity to potentially underestimate the contribution weight of such nodes under strongly heterogeneous data distributions (as our subsequent empirical studies reveal, under the Dirichlet $\alpha = 0.1$ setting, the FPR of such baselines is as high as 21.4%, see Table 3), thereby undermining the original design of federated learning to encompass diverse, broad knowledge systems.

To significantly mitigate the above dilemmas under given edge network heterogeneity constraints, this paper takes a global system perspective, abandons

the isolated evaluation limitations of conventional defenses, and proposes a trust flow-driven trusted federated learning active defense framework. This framework aims to closely coordinate originally scattered verification modules through a “software-hardware intertwined in-depth trust anchor” and a “temporal dynamic evaluation mechanism”. In the underlying physical domain, the system introduces a Trusted Execution Environment (TEE) to establish a hardware admission line of defense, mitigating external identity forgery and unauthorized process tampering risks. In the application algorithm layer, breaking conventional inspection blind spots, we innovatively design an orthogonal state-transition automaton of “historical value flow and multi-dimensional instant risk flow”, thereby ensuring comprehensive broad knowledge convergence while achieving full-lifecycle collaborative defense against various high-dimensional latent threats.

The main contributions of this paper are summarized as follows:

1. **Constructing a dual software-hardware trust anchor and full-lifecycle defense mechanism.** Distinct from single-point defenses that are passively filtered only at the moment of aggregation, this design completely advances the defense depth and permeates the entire model evolution. On the hardware side, it relies on TEE on-chip enclaves to build tamper-resistant physical node admission and runtime environmental fingerprints. On the software side, smoothly inheriting the initial hardware baseline, it realizes a long-term closed-loop management of cross-hardware admission, clean content auditing, temporal reputation calculation, and layer-wise dynamic control.
2. **Proposing an orthogonal dual-stream reputation evolution system adapted to long-term edge environments.** Targeting the core pain point in traditional cumulative reputation mechanisms operating over long cycles—“frequently causing permanent miscalculations of valid long-tail nodes while advanced lurkers evade detection”—this paper pioneers decoupling the identity evaluation logic into two independent, mutually exclusive operating tracks: the “historical performance flow” (*HistPerf*), accumulated through clean contributions, is used to gently preserve the voice and grant pardons to heterogeneous compliant groups (*intuitively, HistPerf acts like a central bank’s personal credit score: it allows compliant nodes to temporarily fall behind due to skewed data, but encourages them to earn back credit pardons through long-term long-tail interactions*); whereas the aggressive “instant risk flow” (*RiskEMA*), aggregating sensitive probes deep in multi-layer neural networks, is exclusively responsible for executing rapid high-risk fuse interceptions (*conversely, RiskEMA works like a one-vote veto security scanner at a boarding gate: once specific poisoning features are triggered, regardless of its high historical contribution, it is decisively blocked at that round*). The system, within designed empirical and theoretical operational bounds (malicious ratio $< 50\%$, Non-IID $\alpha \geq 0.1$), effectively alleviates the trade-off challenge between missing hostile nodes and mistakenly injuring benign ones.

3. **Establishing a clean baseline-based layer-wise heterogeneous aggregation mechanism.** In defending against advanced backdoors, the framework incorporates a core reference origin distilled from a small amount of isolated server data. Combined with the differentiated sensitivity of deep and shallow neural networks to poisoning, it implements risk-gated layer-wise admission, momentum re-calibration of trusted survivors, and fine-grained parameter dynamic L2 clipping, systematically blocking malicious triggers from transparently propagating across deep neurons.
4. **Completing empirical validation and end-to-end platform deployment in multi-morphology security environments.** This paper conducts deep simulation testing utilizing a cross-computing-power edge architecture populated by a 20-client cluster (across 5 independent repeated runs). Facing 6 hybrid attacks and long-tail heterogeneous data distribution environments, the system achieves an exceptionally high and stable malicious node recall performance (TPR) within the configured adversarial intensity range (covering extreme-risk hybrid tests approaching the 50% theoretical limit). It sustains overall collaborative trust with a permanent miskill rate approaching zero ($\text{FPR} \approx 0\%$), secures a main-task average accuracy of 92.31% ($\pm 0.09\%$), and effectively pins the backdoor Attack Success Rate (ASR) expectation at 10.21% ($\pm 0.05\%$). These results validate its deployment resilience as a highly reliable federated engine architecture in practical applications.

2 Background and Related Work

2.1 Synergy Architecture of Federated Learning and Edge Computing

Federated Learning (FL), as a privacy-preserving computing paradigm, allows for the collaborative construction of global knowledge models without data leaving local physical domains through joint parameter adjustments. In edge computing scenarios, a Client-Edge-Cloud collaborative architecture is widely implemented: edge nodes (covering IoT devices and basic computing gateways) bear local computational stress and generate gradients, while the central heterogeneous server is responsible for summarizing, aggregating, and sequentially dispensing global models. However, this architecture naturally engenders extremely severe “system heterogeneity” and “Non-Independent and Identically Distributed (Non-IID) data” traits due to the broad span of communication chains and extreme disparities across edge computing devices. Targeting this architectural bottleneck, cutting-edge explorations have emerged recently, such as Clustered FL [9], adaptive model pruning-expanding (FedPE) [10], and split federated learning frameworks (ParallelSFL) [11]. Nonetheless, as the architecture becomes more complex, such evolutions not only stretch the main convergence timeline but also provide more opportunities for latent infiltration by hostile Byzantine nodes.

2.2 Edge Disguise, Model Poisoning, and Stealthy Backdoor Attacks

During fully open federated wide-area interactions that lack mandatory admission audits, it is difficult for central hubs to effectively verify the legitimacy and gradient purity of local data collection sources. This has led to current malicious attack methods evolving into two deadly high-risk branches on the edge side: one is highly destructive **untargeted poisoning attacks**, such as Sign-flipping and Gradient Scaling, whose purpose is to violently desynchronize global weights and directly execute large-scale Denial of Service (DoS); the second is highly stealthy **targeted backdoor attacks**. Attackers (e.g., via label flipping, clean-label noise injection, deep semantic backdoor poisoning) hide pixel-level triggers within small-scale hidden samples. Normally, they feign high-accuracy contributions to the dominant classification tasks, but when encountering inferences carrying specific patterns, they force the model to direct misclassifications onto triggered samples.

2.3 Mechanisms and Irreconcilable Limitations of Existing Active Defenses

To counter the aforementioned security threats, pioneering defense measures largely adopted geometric statistical filtering mechanisms (such as distance-isolated Krum [4] and dimension-wise median-clipped Trimmed Mean [5]). In the newest frontier research, active trust evaluations based on temporal observation are becoming mainstream: for example, FLTrust’s cosine-guided evaluation based on small batches of clean validation sets [7], FoolsGold’s anti-collusion similarity culling [8], and even adaptive dynamic aggregators integrated at the edge level like RaSA [12]. Meanwhile, explorations focusing on generalized malicious client detection [13], fine-grained trustworthy training framework designs (like TMT-FL) [14], and multi-level adaptive protection architectures [15] have significantly enriched the dimension of purely software-evaluated defenses. In addition, addressing stealthy backdoors and collusion attacks derived from model poisoning, academia has correspondingly proposed FLPurifier based on decoupled contrastive training [16], RoseAgg targeting collusion defense [17], and ShieldFL built on dual-trapdoor mechanisms [18]. However, the aforementioned purely software-based defenses face severe challenges in strong Non-IID and advanced attack scenarios: if the rejection threshold gates are raised, it becomes easy to sweep legitimate nodes housing rare righteous features out of doors (causing FPR to rise significantly). Furthermore, against attackers possessing higher system operation privileges (e.g., Root level), purely software probes, lacking a root of trust anchored in physics, often fail to enact complete defensive closed loops against violations cascading from underlying monitor tampering.

2.4 Trusted Hardware Roots (TEE) in Federated Environments

To cover the physical vacuum left by purely algorithmic control over host environments, researchers have begun leveraging Trusted Execution Environments

(TEEs), such as Intel SGX and ARM TrustZone, to establish high security isolation “on-chip enclaves” within federated participant devices. TEE isolation zones utilize encrypted memory areas to execute classified code; even if an attacker gains underlying Operating System (OS) privileges, it is exceedingly difficult for them to tamper with the stored key attestation and verification signatures (assuming standard security models without addressing advanced side-channel attacks) (as demonstrated by Liao et al.’s tamper-resistant inference mechanism [19] and the RPPFL hardware-encapsulated defense [20]). Recent studies have broadly confirmed the foundational fallback role of TEEs in federated training integrity assurance [21], adversarial attack mitigation [22], and Industrial IoT data distributed trust sharing [23]. Its feasibility for distributed federated deployment scaling under SGX environments [24] and efficient hierarchical concurrent training acceleration strategies [25] have also been successfully demonstrated in the industry. The framework presented in this paper absorbs this purest underlying hardware isolation to thereby construct an intertwined software-hardware defense-in-depth architecture. Distinct from essentially diverging researches like RPPFL [20], which only leverages TEEs for static gradient encryption mechanisms, or TMT-FL [14], which singularly focuses on backend anomaly evaluation, this paper—to the best of our knowledge—is the first to temporally cross-couple underlying remote hardware attestations with top-level dual-stream orthogonal dynamic reputation evolution in edge federated learning scenarios: not only verifying parameters from the system kernel (cloud) but also attaching soft-anchored locks (TMAA) possessing reverse-monitoring capabilities and highly credible evidentiary reporting mechanisms onto the edge through hardware hands, achieving a millisecond-level temporal synergy between cross-hardware mandatory admission rules and software dynamic risk fuse interventions.

3 System Model and Problem Definition

3.1 Client-Edge-Cloud Synergy Architecture and System Flow Modeling

In the “Trust Flow” defensive logical grid designed in this paper, the system operates within a typical large-scale Client-Edge-Cloud un-trusted communication link environment (the global cooperative physical architecture is shown in Fig. 1). The entire operational architecture comprises an overriding **Parameter Aggregation Audit and Verification Center (Server, S)** at the top, and a constellation of K loosely connected **Edge Node Container Ensembles** $\mathcal{C} = \{c_1, c_2, \dots, c_K\}$ wandering in the shallow network layers, which connect intermittently and are potentially susceptible to becoming controlled poisoners at any moment.

The system initially establishes TEE encrypted channels within the kernel of each edge device c_k by deploying hard access credentials, and subsequently executes its joint modeling objective: minimizing the expected empirical risk

over the aggregate data distributions $\mathcal{D}_k$ across all physical edge entities:

$$\min_W \mathcal{L}(W) = \sum_{k=1}^K p_k \mathcal{L}_k(W) \quad (1)$$

where $p_k = |\mathcal{D}_k| / \sum |\mathcal{D}_j|$. At the dawn of the t -th training round, the server broadcasts a baseline model replica $W_{global}^{(t-1)}$. The secure agent nodes (starting from $W_{k,0}^{(t)} \equiv W_{global}^{(t-1)}$) perform E rounds of local Stochastic Gradient Descent (SGD) computations using a localized learning rate η_l over their Non-IID datasets. The resulting localized parameter increment can be expressed as an accumulative tensor shift:

$$\Delta W_k^{(t)} = W_{k,E}^{(t)} - W_{global}^{(t-1)}, \quad \text{where } W_{k,e}^{(t)} = W_{k,e-1}^{(t)} - \eta_l \nabla \mathbb{E}[\ell(W_{k,e-1}^{(t)}; x, y)] \quad (2)$$

where the expectation $\mathbb{E}[\cdot]$ denotes the mathematical expectation sampled via randomized mini-batches over the local dataset $\mathcal{D}_k$ (the batch size is uniformly set to 64 in the simulation framework configuration of this study).

Finally, legitimate increments along with their ‘‘Hardware-level Anti-tampering Attestation Quotes’’ endorsed by TEE signatures are uploaded simultaneously into the risk filtration network, allowing the central server to implement robust updates underpinned by a multi-stream evolutionary trust-scoring system:

$$W_{global}^{(t)} = W_{global}^{(t-1)} + \text{Agg}(\{\Delta W_k^{(t)}\}_{k \in \Phi}) \quad (3)$$

where $\Phi \subseteq \mathcal{C}$ encapsulates the ensemble of valid clients surviving global consensus through the current round; while $\text{Agg}(\cdot)$ epitomizes the core secure trust aggregation operator engineered in this paper. Its deterministic adaptive fusion mechanisms decoupled upon abiding historical scores and high-dimensional instant risks will be systematically unfurled in Section 4.

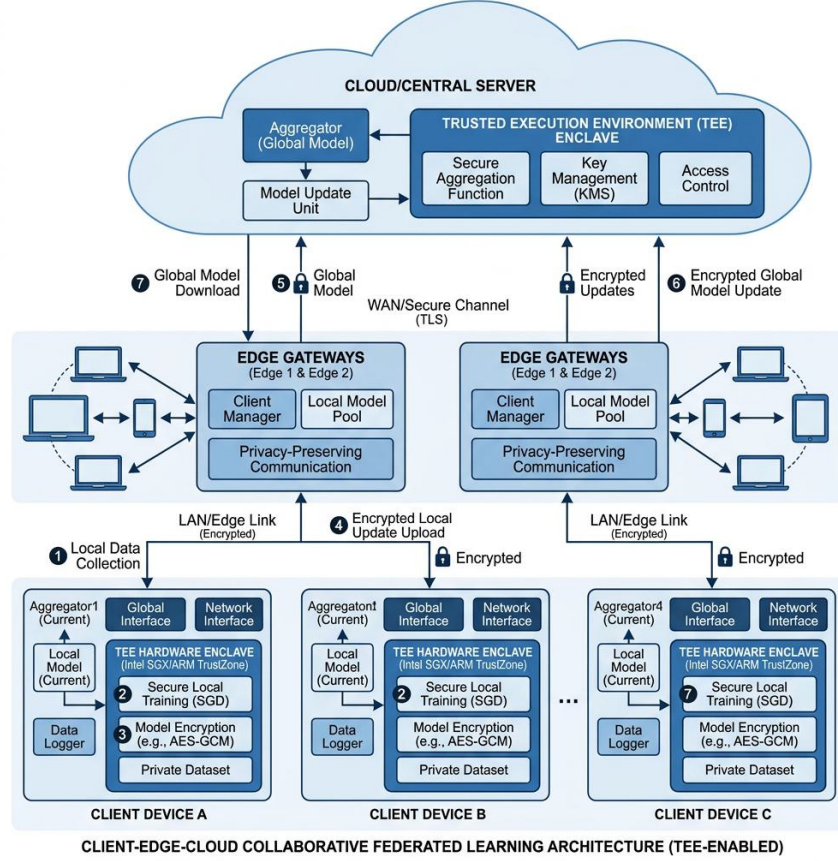


Fig. 1. Federated Computational Network Interaction: Client-Edge-Cloud collaborative physical architecture diagram encrypted upon TEE anchor points.

3.2 Composite Threat Assumption and Strong Trust Fundamental Boundary (Threat Model)

Any pre-established security declaration must demarcate boundaries of disparate adversarial and defensive capabilities and foundation assumptions. To fully extract the framework’s potent capabilities, this manuscript delineates three stratifications of preventive virtual entities alongside bottom-tier trust red lines (the multi-dimensional attack surfaces under generalized edge networks are visually summarized in Fig. 2):

- **Hardware Isolation Security Boundary Assumption (The Cornerstone of Trust):** The system is intrinsically rooted within limits verifiable through hardware bounding; that is, underneath standard logical security modeling assumptions (setting aside prerequisites exploring micro-architectural temporal tracking against silicon arrays, localized side-channel

analytic inferences, or physical brute-force probing mechanisms [20]), it is presumed that even if the external Host OS endures an escalation of kernel-level subversions and total control, the inner-resilient hardware privilege zeniths (e.g., TEE verification jurisdictions constituted by TrustZone/SGX architectures) can persistently sustain elevated computational-difficulty isolation levels. This offers fundamental grounding for issuing anti-forgery runtime behavioral properties that cannot be effortlessly bypassed or unethically denied at low thresholds.

- **Honest-but-curious Cloud Aggregation Vault:** This warrants that our deployed dual-stream reputation integration pools and multi-tiered multiplier functions will be stringently followed and deterministically executed without deviation; it simultaneously defends against actions encompassing unauthorized access to sensitive raw albums via plaintext interpretations.

- **Omnipresent and Meticulously Camouflaged Internal Colluding Factions (Internal Attackers/Poisoners):** Distinguished from external passive attackers who can only snoop on generic datagrams without initiating validated certifiable handshakes, it is structurally assumed our hypothetical adversaries have victoriously nestled inside arrays of legitimate devices armed with pristine admission credentials (carrying an upper boundary warning ratio $< 50\%$). The internal disruptors possess virtually unfettered authority to dynamically tamper local label paradigms, infuse inherently corrupted data points (structuring noisy clean-label toxic imageries), or brutally wrench underlying PyTorch SGD increment step matrices to coerce aggressive sign inversions and massive magnification (gradient parameter scaling anomalies). We emphatically note, albeit theoretic tolerance brackets scale firmly to a 50% hostile fraction, realistically encompassing configurations position foundational benchmarks toward prevalent high-intensity empirical 30% limits. The extended deep-water stress tests directly encroaching the 50% barrier stand formally chronicled as autonomous verifications mapping system resiliency boundaries.

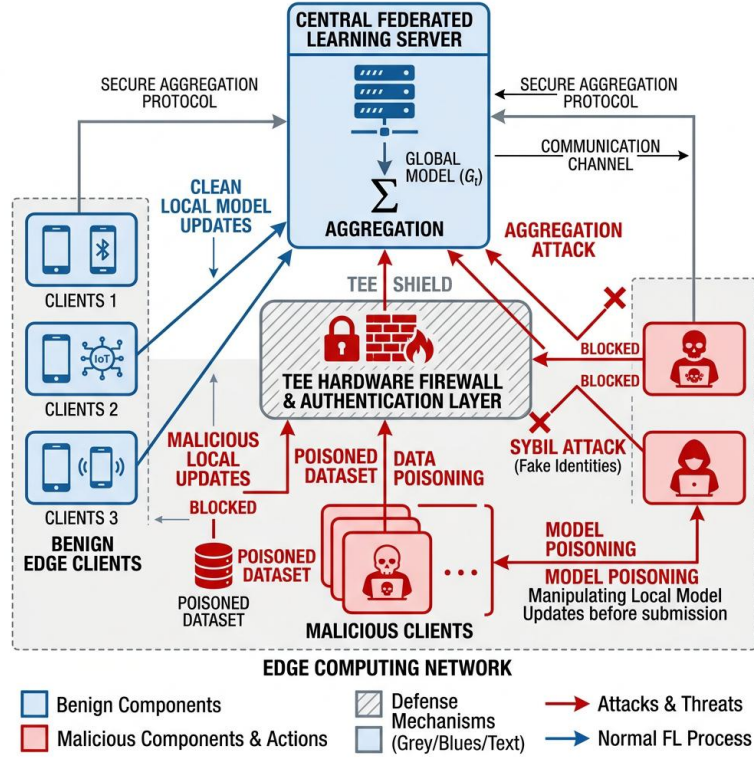


Fig. 2. Security penetration and multi-dimensional composite poisoning threat modeling schematics within wide-area edge topographies.

3.3 Security Architecture Core Defensive Optimization Design Goals

Standing amidst extreme data dispersion and cross-pressured from a 30% up to an agonizing 50% malicious penetration jeopardy, the systemic matrix charted within this document aims to shatter three intricately formidable bounds constraining active defenses:

1. **Supreme True-Positive Filtering Capability and Sturdy Mainstay Convergence Network:** Regardless of how agonizingly protracted the cyclic durations or minuscule the deviating displacement angles internal camouflaged poisoners coordinate via collective manipulations, the system shall decisively pinpoint and purge malignant actors. Specifically, maintaining global macro-convergence paths inherently accomplishing virtually 100% precise true-positive malicious recall rates, all while suppressing posterior Backdoor ASR ceilings entirely avoiding security chaos instatement.
2. **Unwavering Non-Disconnection Defense over Compliant Heterogeneous Segments:** Overcoming persistent mistaken-kill paradigms strictly

dictated by pure geometric coordinates or flat threshold slashing methodologies—allowing compact compliant machineries carrying wildly angled localized statistics due strictly to regional datasets to suffer minimal short-term blockage; however, formally granting these deeply skewed yet functionally legitimate cohorts capabilities upholding participative purposes securely guided through extensive regenerative mechanisms governed via integrated evolutionary tolerances (aiming for absolute minimal irreversible misclassification, achieving long-term False Positive Rate (FPR) nullification).

3. **Economical Spatio-temporal Fluid Execution and Minimal Overhead Expansion:** Relinquishing solutions mandating monolithic hundred-fold global delay multipliers commonly originating in Fully Homomorphic Encryption validations or convoluted Zero-Knowledge authentications purposed towards mitigating poisoning; instead systematically funneling cardinal computing expenditures explicitly into undeniable hardware characteristic fingerprinting interwoven dynamically with orthogonal-dimensional array tracking, fulfilling acute operational limits expected from heavily constrained edge LXC/Docker topologies.

4 Proposed Method: Trust Flow-Driven Trusted Federated Framework

Addressing the aforementioned composite threat model, this section meticulously details the introduced “Trust Flow-Driven Trusted Federated Learning” defense architecture. The foundational philosophy consists of discretizing an entire algorithmic federated epoch into five synergistic pivotal phases: pre-training hardware admission, intrinsic behavioral monitoring, pre-aggregation content surveillance, orthogonal state evolutionary transition, and finally, layer-wise disparate dynamic aggregation. Divergently forsaking solitary isolated point-defense methodologies, the architectural premise maneuvers by transmitting and tracking holistic **Trust Flows** interlocking across these five continuous chapters, curating an enduring persistent regulatory loop.

For seamless subsequent mathematical formulation and algorithmic proofs, Table 1 delineates the standard definitional configurations representing the cardinal systemic components across this work’s fundamental expressions.

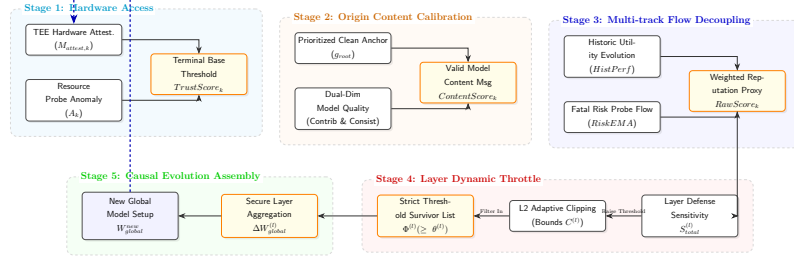
4.1 Phase 1: Hardware-anchored Client Admission and Initial Trust Scoring

To proactively eliminate “Sybil nodes” which casually manipulate identities, alongside highly compromised machineries operating out of drastically breached environments at the very origin locus, the structural configuration maneuvers Trusted Execution Environment (TEE) tech elements integrating a **Trusted Management Attestation Agent (TMAA)**. Prior to successive learning rounds, operating clients are strictly obligated to relay a rigorous Remote Attestation Quote—containing chain evaluations mapping Platform Configuration

Table 1. Core System Parameters and Trust Flow Symbolic Notations.

Symbol	Definitional Description	Symbol	Definitional Description
$W_{global}^{(t)}$	Global model parameter weights at round t	$\mathcal{A}$	Active node subset h
ΔW_k	Model gradient/parameter increments by client k	$\Phi^{(l)}$	Effectively secure sur
$TrustScore_k$	TEE hardware-anchored foundational admission trust	$HistPerf_k$	Historical performanc
$ContentScore_k$	Current-epoch content quality scoring measured dual-ways	$RiskEM A_k$	Instantaneous risk co
$RawScore_k$	Unifying tri-dimensional score matrix applied with risk discounting	$M_{attest,k}$	Structural Hardware
g_{root}	Absolute pristine validation baseline established via server bounds	A_k	Absolute physical res

Feed Global Knowledge and Reputation back to Edge Terminals (Closed-loop Starts)

**Fig. 3.** Holistic Architectural Outline of the Trust Flow-Driven Five-stage Closed-Loop Defense.

Registers (PCRs)—derived directly from in-exportable Hardware Device Unique Keys (DUKs) strictly aimed at the central governance. Permitting successful penetration validation matching system integrity restrictions translates the verification flag assignment $M_{attest,k} = 1$ (otherwise it automatically locks at 0, culminating in absolute explicit blockage denial).

Extrapolating strictly governed metrics logged exclusively by the TMAA surveillance loop (e.g., CPU contextual paradigms alongside algorithmic process legality mapping), the server systematically projects an innate anomaly deviation fingerprint signature evaluating $A_k \in [0, \infty)$ for nodes triumphantly overcoming gate entry. Consequently, an initial functional access trust allowance $TrustScore_k \in [0, 1]$ dedicated for intra-round examination mechanisms is computationally extrapolated:

$$TrustScore_k = M_{attest,k} \cdot \exp(-\lambda \cdot \max(0, A_k - \tau)^\rho), \quad (4)$$

where structural control parameter λ enforces absolute gradient steepness dictating penalizations correlated with surging anomalistic deviations; τ mimics bounded tolerance zones forgiving typical physical variance unaligned structurally with aggressive operations; while ρ administers rapid compounding decay accelerations (within default empirical experiment layouts, limits are formulated as $\lambda = 5.0$, $\tau = 0.1$, $\rho = 2.0$ respectively, specific component combinations map-

ping out broader parametric boundaries are broadly enumerated across details listed within Section 5).

4.2 Phase 2: Server-prioritized Reference Direction Construction and Isomorphic Auditing

Conducting solitary cosine metric cross-assessments directly amidst client gradients frequently incites unmitigated vulnerability against organized aggregations forming collusive adversarial networks. Emphasizing rigid mitigation, this paradigm pioneers the **Vanilla Reference Prioritization** standard. Commencing operations, provided the macro-system manages an exceedingly minimal quantity of pristine separated observations (Pure Clean Set), the infrastructure extracts precise gradient trajectory mapping generated natively over top-level global frameworks, formulating g_{root_clean} . Whenever absolute data validation conditions remain categorically inaccessible, the architecture gracefully defaults towards adopting a robust momentum baseline defined through weight-averaging gradients mapped by trusted clean-cache components persisting explicitly through consecutive prior epochs: formally formulated as g_{ref} . Addressing structural distribution arrays, abandoning conventionally biased uniform volume divisions completely, dynamically regulated non-linear coefficients controlled coordinately between TrustScore foundations together with the antecedent cycle’s accumulative risks (RiskEMA gating paths) orchestrate calculations:

$$\omega_k^{ref} = TrustScore_k \cdot (1 - RiskEMA_k^{prev})^2. \quad (5)$$

The absolute unifying global referential orientation guiding the ensuing t -th aggregation maneuver consequently maps to:

$$g_{root} = \begin{cases} g_{root_clean}, & \text{if deterministic verified server gradient boundaries natively persist,} \\ \sum_k \frac{\omega_k^{ref}}{\sum_j \omega_j^{ref}} \Delta W_k, & \text{otherwise.} \end{cases} \quad (6)$$

After attaining the concrete referential vector g_{root} , the governing hub actively evaluates a composite “Content Quality Rating” scaling the absolute parameter deviations contributed via ΔW_k . Systemic computations meticulously calibrate a non-symmetrical harmonic merging integrating Contribution Scaling (dictated entirely by dimensional alignment facing undisputed primary clean root directives) combined symmetrically with Group Consistency Profiling (evaluating vector variance distributions aligning closely clustered trusted majority pools) towards calculating absolute current epoch limits defined functionally as

$ContentScore_k$:

$$S_{contrib,k} = \max(0, \alpha_1 + \alpha_2 \cdot \cos(\Delta W_k, g_{root})), \quad (7)$$

$$S_{consist,k} = \frac{\sum_{j \neq k} TrustScore_j \cdot \max(0, \alpha_1 + \alpha_2 \cdot \cos(\Delta W_k, \Delta W_j))}{\sum_{j \neq k} TrustScore_j + \varepsilon}, \quad (8)$$

$$ContentScore_k = \frac{(1 + \beta_{fusion}^2) \cdot S_{consist,k} \cdot S_{contrib,k}}{\beta_{fusion}^2 \cdot S_{consist,k} + S_{contrib,k} + \varepsilon}. \quad (9)$$

Within the preceding configurations, variables α_1 alongside α_2 (adhering strictly to normalization requirements defining $\alpha_1 + \alpha_2 = 1.0$) facilitate smoothly shifting calibration scales tracking cosine intervals efficiently; whereas $\beta_{fusion} > 1.0$ (calibrated securely empirically around $\beta_{fusion} = 2.0$) symbolizes structurally deterministic preferential anchors representing overriding inclinations commanding the absolute directive orientations (closer proximities converging pure baselines) outclass abstract group-based cooperative continuities. Prioritizing centralized ground truth establishes resilient countermeasures blocking heavily camouflaged conspiratorial alliances while strictly enforcing valid universal convergence pathings.

4.3 Phase 3: Orthogonal State Evolution of Historical Performance Flow and Instant Risk Flow

Approaches clinging universally onto conventional “lowest-ranked blacklisting cuts” chronically manifest profound systemic flaws mapping heterogeneous constraints: typically, whenever legitimate elements structurally trapped retaining profoundly uncommon disparate characteristics invariably gather systematically minimal evaluations persistently through longitudinal tracking epochs, they inherently confront unwarranted execution eradication. The intrinsic developmental pivot crafted mathematically inside this domain involves actively executing absolute “dimensional decoupling” disassembling chronological evolutionary metrics onto twin opposing and firmly distinct temporal progression tracks: deploying the ****Historical Performance Flow (*HistPerf*)**** devoted primarily managing resilient permission threshold contentions, structurally distanced functionally against the rigorously determinative autonomous elimination mechanism regulated fully managed alongside the ****Instantaneous Risk Containment Flow (*RiskEMA*)****.

(1) Historical Performance Flow (*HistPerf*) functions systematically carrying structural evaluation spanning elongated cyclical boundaries. Critically, identifying *HistPerf* strictly precludes any capabilities authorizing permanent “rejection or penalization” terminations. Progressing round by round, utilizing standard variance distributions (z scores inherently extracted) mapped via individual discrete array $ContentScore_k$ rankings competing natively across comprehensive populations, smoothing algorithmic constraints alongside bottom-

Algorithm 1 Admissions and Content Auditing Grounded on Hardware Anchors and Pristine References

Input: Set of structural clients $\mathcal{C}$, received localized dimensional derivations $\{\Delta W_k\}_{k \in \mathcal{C}}$, previous historical boundary states $RiskEMA^{(t-1)}$

Output: $TrustScore_k$, $ContentScore_k$, definitive reliable references defining g_{root}

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1: Stage 1: Hardware-anchored structural validations
2: for every singular localized computing node  $k \in \mathcal{C}$  do
3:   if  $M_{attest,k} == 1$  (TEE physical verifications formally accepted) then
4:     Obtain anomaly structural bounds defining  $A_k$ 
5:     Trust baseline extrapolation  $TrustScore_k \leftarrow \exp(-\lambda \max(0, A_k - \tau)^p)$ 
6:   else
7:     Categorical access rejection executed:  $TrustScore_k \leftarrow 0$ 
8:   end if
9: end for
10: Stage 2: Reference Directives alongside Isomorphic Auditing
11: if Possessing intrinsically validated pure centralized analytical data arrays then
12:   Extraction defining absolute truth pathings dictating  $g_{root} \leftarrow g_{root\_clean}$ 
13: else
14:   Evaluate non-linear distribution mappings  $\omega_k^{ref} \leftarrow TrustScore_k \cdot (1 - RiskEMA_k^{(t-1)})^2$ 
15:   Establishing robust collective trajectory limits  $g_{root} \leftarrow \sum_k (\omega_k^{ref} \Delta W_k) / \sum_j \omega_j^{ref}$ 
16: end if
17: for Functionally certified active components  $k$  do
18:   Global utility orientation limits  $S_{contrib,k} \leftarrow \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, g_{root}))$ 
19:   Mass collaborative integrity values  $S_{consist,k} \leftarrow \frac{\sum_{j \neq k} TrustScore_j \max(0, \alpha_1 + \alpha_2 \cos(\Delta W_k, \Delta W_j))}{\sum_{j \neq k} TrustScore_j + \epsilon}$ 
20:   Harmonic unifying evaluation mapping  $ContentScore_k \leftarrow \frac{(1 + \beta_{fusion}^2) S_{consist,k} S_{contrib,k}}{\beta_{fusion}^2 S_{consist,k} + S_{contrib,k} + \epsilon}$ 
21: end for
22: return  $\{TrustScore_k, ContentScore_k, g_{root}\}$ 

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threshold limiters dictate absolute variable increment limits designated $Signal_k^{upd}$:

$$z_k = \frac{ContentScore_k - \mu_{content}}{\sigma_{content} + \varepsilon}, \quad Signal_k^{rel} = \sigma(z_k), \quad (10)$$

$$Signal_k^{abs} = \text{clip} \left(\frac{ContentScore_k - b}{s}, 0, 1 \right), \quad (11)$$

$$Signal_k^{upd} = \gamma_{rel} \cdot Signal_k^{rel} + \gamma_{abs} \cdot Signal_k^{abs}, \quad (12)$$

$$HistPerf_k^{(t)} = \beta_h \cdot HistPerf_k^{(t-1)} + (1 - \beta_h) \cdot Signal_k^{upd}. \quad (13)$$

Functionally, bounds structured by variables b, s control mapping constraints representing foundational normalizing boundaries dictating fundamental minimal operational standards; components γ_{rel} alongside γ_{abs} (enforcing absolute cumulative combinations aggregating 1.0) adjust balancing weightings dynamically shifting preferences separating rigorous internal contestation models against standardized universally fixed expectations. During situational moments wherein evaluations driving $HistPerf$ cascade exceedingly poorly, corresponding elements forcefully transition merely into highly restrictive temporary placements formally titled ‘‘Historical Soft Isolation Zones’’ (entirely forbidding structural incorporation representing current aggregations while strictly enforcing unbarred continual presence mapping the macro-environment domains). Such architecture successfully shields under-represented heterogenic elements.

(2) Instantaneous Risk Containment Flow (*RiskEMA*), conversely acts systematically representing the absolute cornerstone eliminating explicitly aggressive hostility nodes. Conclusively circumventing simplistic angular metric dependencies targeting baseline matching calculations, architecture assimilates comprehensively fused multi-dimensional analytic components deriving profoundly sensitive parametric data indicators extending significantly across deep networks (such tracking elements comprising loss oscillatory turbulence mapping r_{probe} , layered neuronal polarity variance configurations r_{grad} , embedded stealth trigger mechanisms $r_{trigger}$, inherently integrated beside pixel-structured preliminary scans r_{pixel}). The apparatus continuously extracts maximum absolute extremity occurrences spanning explicitly across unified cross-domain probe risk assessments capturing instantaneous limits:

$$Risk_k^{inst} = \max\{r_{report}, r_{grad}, r_{probe}, r_{pixel}, r_{trigger}, r_{sign}, r_{peer}, \dots\}, \quad (14)$$

swiftly triggering retrogressive algorithmic exponential decay combinations (establishing temporal tracking boundaries mapped through structural momentum configurations fixing $\beta_r = 0.85$ defining dynamic halving limitations):

$$RiskEMA_k^{(t)} = \beta_r \cdot RiskEMA_k^{(t-1)} + (1 - \beta_r) \cdot Risk_k^{inst}. \quad (15)$$

Tightly intertwining the previously identified high-dimensional risk variables parallel against historical utility metrics generating robustly distinct orthogonal mathematical extrapolations, a complete secure mapping timeline dictates an all-encompassing logical tracking mechanism formulated as explicitly detailed

security lifecycle phase automations (diagrammatic topological representations documented structurally via Fig. 4), universally outlining strict four-tier regulatory classification bounds spanning completely across all interconnected network participants:

- **Normal Standard Elements (NORMAL)**: Universally integrated completely representing unhindered current analytical combination deployments. Upon the precise instance where intrinsic tracking methodologies explicitly record disruptive escalations (triggering sharp RiskEMA trajectories) or alternately observe progressively plunging historical efficacy combinations, respective operational units are automatically categorized shifting unilaterally rightwards towards Suspect delineations representing explicitly segregated soft isolation statuses mappings.
- **Highly Flagged Suspicious Components (SUSPECT)**: Entities systematically activating alarm protocols directly incur reinforced localized inspections mapping heavily restricted functional observation channels. Conditioned on subsequent measurements actively displaying verified diminishing risk variables representing sustained compliance standards tracking normalized pathways, the platform automatically facilitates unconditional administrative reviver directives reverting capabilities safely towards establishing Normal bounds. Contrarily, manifesting continued deliberate limit threshold exceedances definitively seals administrative decisions relocating corresponding factors irrevocably across complete isolation protocols.
- **Absolute Structural Isolation Frameworks (QUARANTINE)**: Entirely abolishes absolute dimensional inclusions mapping globally consolidated core parametric influences, thereby explicitly restricting capabilities allowing contributions purely mapping solely alongside non-consequential secondary verification computational loads. Demonstrating verifiable subsequent cyclic compliant tracking (triggering conditional relief mappings) re-establishes pathways back establishing verified Normal standards; alternatively, accumulating historically undeniable extensive penalization sequences formally breaching systemic lock-up rules definitively accelerates absolute downfall mechanics.
- **Permanent Absolute Revocation Blacklist Frameworks (BLACKLIST)**: Enforcing unilaterally absolute irreversible ban configurations. Successfully engaging hardware algorithmic limits universally severing explicit core structural handshake authentication channels dictated by deep TEE verification pathways permanently stripping all future collaborative certification pathways completely.

This highly implicit algorithmic tracking methodology universally integrating exceptionally extensive historic accommodative flexibilities (implementing broad administrative pardon structures) explicitly grants the macro-framework capabilities aggressively executing highly disjointed heterogeneous parameter classifications meticulously preserving absolute capabilities driving unyielding limit-approaching zeros representing comprehensive operational miscalculation metrics.

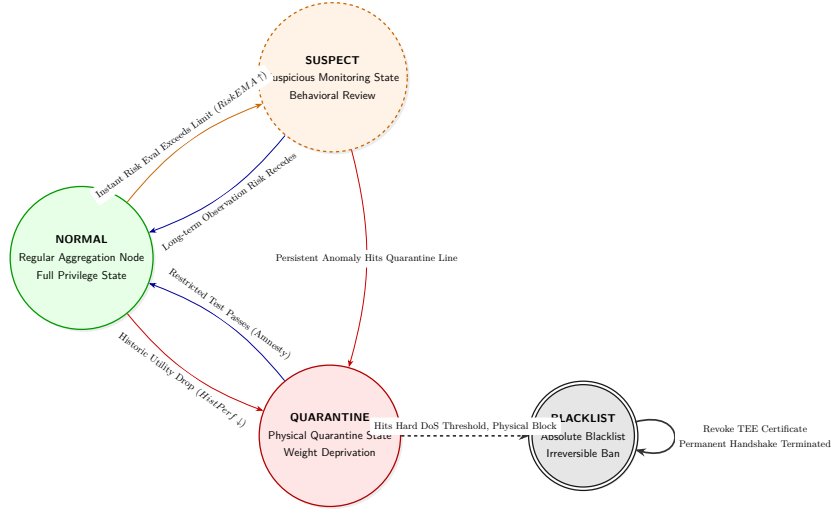


Fig. 4. Logical automation diagram defining four-state transition supervision tracking integrating cross-disciplinary client historical accumulation juxtaposed besides localized burst risks.

Cumulatively, *TrustScore* (baseline hardware credibility mappings), *ContentScore* (immediate structural contribution measurements evaluating present periods), and *HistPerf* (historical reputation foundations) synergize explicitly determining aggregate determinative index scaling evaluating the ultimate uninfluenced current interval authority limits, subsequently layering absolute risk restriction multipliers definitively extrapolating absolute bounds mapping $RawScore_k$:

$$RawScore_k^{base} = (TrustScore_k)^\alpha \cdot (ContentScore_k)^\beta \cdot (HistPerf_k^{(t-1)})^\gamma, \quad (16)$$

$$RawScore_k = RawScore_k^{base} \cdot (1 - RiskEMA_k^{prev})^p, \quad (17)$$

within which predefined hyperparametric exponential modifiers mapped by variables configuring α, β, γ, p systematically structure corresponding authoritative limit controlling strengths encompassing prior-referenced primary functional measurement bounds directly allocating absolute authority measurements dictating feedback combination priorities (where optimal variable governing arrays similarly parallel experimental hyperparameter boundary specifications natively cataloged through successive experimental domains).

4.4 Phase 4: Risk-gated Layer-wise Differentiated Robust Aggregation

Structurally sophisticated backdoor manipulation trajectories established by highly covert adversarial bodies traditionally systematically remain deeply entrenched mapping deeply embedded explicitly isolated neuronal mapping combinations.

Algorithm 2 Dual-stream Orthogonal Trust Environment Evolutionary Formulations (HistPerf & RiskEMA)

Input: $ContentScore_k$, encompassing comprehensive high-dimensional probe limits $\{r_{probe}, r_{grad}, \dots\}$

Output: Derived cumulative historical markers $HistPerf_k^{(t)}$, transient cyclic bounds $RiskEMA_k^{(t)}$, alongside comprehensive global authority calculations representing $RawScore_k$

- 1: **Trajectory A: Accumulative Utility Historical Flow Operations (Managing Elastic Allowances)**
 - 2: Tabulate comprehensively integrated scoring means computing $\mu_{content}$ combined against global standard variations mapping $\sigma_{content}$
 - 3: **for** Functionally certified active components encompassing k **do**
 - 4: Calculate corresponding relative variance competitiveness components mapping $Signal_k^{rel} \leftarrow \sigma((ContentScore_k - \mu_{content})/(\sigma_{content} + \varepsilon))$
 - 5: Synthesize independent structured baseline combinations representing explicit bounds configured via $Signal_k^{abs} \leftarrow \text{clip}((ContentScore_k - b)/s, 0, 1)$
 - 6: Evaluate structurally derived updates defining components configuring $Signal_k^{upd} \leftarrow \gamma_{rel}Signal_k^{rel} + \gamma_{abs}Signal_k^{abs}$
 - 7: Refresh dimensional array vectors computing updated states specifying $HistPerf_k^{(t)} \leftarrow \beta_h HistPerf_k^{(t-1)} + (1 - \beta_h)Signal_k^{upd}$
 - 8: **end for**
 - 9: **Trajectory B: Instant Risk Extinguishment Extrapolations (Definitive Veto Configurations)**
 - 10: **for** Each individual distinct operational identity mapping k **do**
 - 11: Systematically calculate singular distinct transient bounds evaluating extremity paths $Risk_k^{inst} \leftarrow \max\{r_{report}, r_{grad}, r_{probe}, r_{pixel}, r_{trigger}\}$
 - 12: Restrict arrays utilizing progressively updated chronological exponential trajectories establishing limits defining $RiskEMA_k^{(t)} \leftarrow \beta_r RiskEMA_k^{(t-1)} + (1 - \beta_r)Risk_k^{inst}$
 - 13: **if** Structural measurements exceed baseline bounds mapped explicitly beyond isolated limits dictating $RiskEMA_k^{(t)} > \text{Isolation Boundary thresholds}$ **then**
 - 14: Force administrative allocation shifting identity metrics persistently towards structural suppression $RiskIsolated \leftarrow RiskIsolated \cup \{k\}$
 - 15: **end if**
 - 16: **end for**
 - 17: **Absolute Unifying Authority Algorithmic Computations**
 - 18: **for** Functionally unrestricted participants categorized implicitly $k \notin RiskIsolated$ **do**
 - 19: Accumulate foundational baseline configurations detailing combined scores specifying $RawScore_k \leftarrow (TrustScore_k)^\alpha (ContentScore_k)^\beta (HistPerf_k^{(t-1)})^\gamma$
 - 20: Assimilate structural penalization combinations strictly enforcing restricted discounting measurements $RawScore_k \leftarrow RawScore_k \cdot (1 - RiskEMA_k^{(t-1)})^p$
 - 21: **end for**
 - 22: **return** Combined matrix outcomes mapped explicitly recording $\{HistPerf_k^{(t)}, RiskEMA_k^{(t)}, RawScore_k\}$
-

Consequently, bluntly enforcing absolute universal multiplicative aggregating unifications extending entirely spanning complete macroscopic model parameters intrinsically invites distinct capabilities explicitly enabling structurally latent infiltrations navigating inherently propagating directly inside unified parameter topologies. Subsequently, this methodology introduces specialized structural admission configurations actively implementing uniquely restricted combinations executing uniquely mapping explicitly differentiated (Layer-wise) validation restrictions actively structured systematically parallel tracking individualized architectural network depth delineations determining bounds mapping l .

Principally constituting primary configurations, executing rigorous deterministic evaluations categorizing completely authorized subsets filtering entirely active functional components encompassing parameters implicitly determining active participants $\mathcal{A}$:

$$\mathcal{A} = \{k \mid k \notin RiskIsolated \wedge k \notin HistSoftIsolated\}. \quad (18)$$

Subsequently transitioning through sequential execution bounds evaluating corresponding dimensional mappings aligned structurally targeting layer configuration l , systematic categorizations compute individual localized security requirement stringencies. Topologically shallow neural structures traditionally predominantly retain standardized public attributes broadly mapping structurally universal physical visual properties, whilst intrinsically complex deep networks fundamentally represent profound architectural deviations dictating fundamental functional representations (mapping ultimate generalized structural utility viability limits) structurally creating distinct frameworks profoundly attractive capturing deeply integrated vulnerability exploitation characteristics. Exploiting three-dimensional independent evaluating logic arrays dynamically measuring structural mapping parameter variance variables establishes absolute combined foundational tracking limitations computing uniquely scaled component sensitivity values structuring $S_{total}^{(l)}$:

$$S_{privacy}^{(l)} = \exp\left(-\tau_p \frac{l}{L}\right), \quad (19)$$

$$S_{utility}^{(l)} = \frac{\|g_{ref}^{(l)}\|_2}{\max_m \|g_{ref}^{(m)}\|_2 + \varepsilon}, \quad (20)$$

$$S_{security}^{(l)} = 1 - \frac{\sum_{k \in \mathcal{A}} TrustScore_k \cdot \cos(\Delta W_k^{(l)}, g_{ref}^{(l)})}{\sum_{k \in \mathcal{A}} TrustScore_k + \varepsilon}, \quad (21)$$

$$S_{total}^{(l)} = w_p S_{privacy}^{(l)} + w_u S_{utility}^{(l)} + w_s S_{security}^{(l)}. \quad (22)$$

Structurally delineating variables specifying w_p, w_u, w_s implicitly characterize entirely distinct algorithmic coefficients completely integrating normalizing requirement delineations capturing distinct structural operational necessities. Formulations registering exceptionally commanding cumulative evaluations indicating extreme unified limits mapped across comprehensive variables analyz-

ing boundaries mapping $S_{total}^{(l)}$ structurally unequivocally determine corresponding localized framework levels intrinsically representing exceptionally susceptible configurations experiencing heavily corrupted malicious contamination topologies; comprehensively resulting structurally prompting direct structural restrictions actively amplifying base operational filtering limitations structuring progressively restrictive boundaries mapping limits $\theta^{(l)} = \mu_{base} + \lambda_s \cdot S_{total}^{(l)}$ seamlessly executed conjunctly aggressively restricting baseline capabilities compressing maximal mathematical dimensional bounds computing explicitly limited L2 functional parameter configurations restricting values establishing bounds scaling effectively $C^{(l)} = C_{base}/(S_{total}^{(l)} + \varepsilon_c)$.

Sequentially proceeding corresponding configurations defining explicit third-stage architectural deployments, strictly only distinctive explicitly operational variables traversing baseline authoritative minimum requirements evaluated comprehensively mapping parameter outputs establishing baseline thresholds where currently evaluated combinations satisfy functional limits represented through $RawScore_k$ distinctively bypassing baseline metrics determining values strictly expanding beyond restrictive thresholds explicitly $\theta^{(l)}$ successfully systematically constitute exclusively admitted configurations comprehensively mapping entirely validated “cleared surviving entities” specifically recognized bounding explicitly towards structural layer elements categorized $\Phi^{(l)}$ functionally allowed structurally mapping explicitly towards parameter submission pipelines. Completing transitions effectively encompassing successfully preserved validation lists subsequently strictly prompts independent normalization operations executing systematically independent authoritative fraction capabilities uniquely specifying independent component evaluation delineations mapped $\tilde{w}_k^{(l)}$ dynamically executed comprehensively establishing forced systematic L2 clipping boundaries directly truncating anomalous parameter vibration deviations mapping corresponding variables specifically dictating distinct variable bounds (mapped configuration $scale_k^{(l)}$):

$$\Phi^{(l)} = \{k \in \mathcal{A} \mid RawScore_k \geq \theta^{(l)}\}, \quad \tilde{w}_k^{(l)} = \frac{RawScore_k}{\sum_{j \in \Phi^{(l)}} RawScore_j + \varepsilon}, \quad (23)$$

$$scale_k^{(l)} = \max \left(1, \frac{\|\Delta W_k^{(l)}\|_2}{C^{(l)}} \right), \quad \widehat{\Delta W}_k^{(l)} = \frac{\Delta W_k^{(l)}}{scale_k^{(l)}}. \quad (24)$$

Finalizing operations tracking distinctly mapped fourth-phase independent evaluation arrays, each uniquely restricted comprehensively limited individually structural combination progressively compiles dynamically independent restricted fractional allocations integrating functionally individual layer enhancements establishing comprehensive variables explicitly outlining explicitly:

$$\Delta W_{global}^{(l)} = \sum_{k \in \Phi^{(l)}} \tilde{w}_k^{(l)} \cdot \widehat{\Delta W}_k^{(l)}, \quad (25)$$

Algorithm 3 Layer-wise Differentiated Aggregations and Dynamic Clipping
 Driven by Risk-gated Methodologies

Input: Set representing verified operational components mapping $\mathcal{A}$, array tracking fundamental calculations determining metrics computing $RawScore_k$, individually evaluated hierarchical structural parameter increments formatting values detailing $\{\Delta W_k^{(l)}\}$, combined completely comprehensive functional layer limits representing variables structurally mapping network depths tracking limits evaluating L

Output: Integrated thoroughly verified completely evaluated completely secured entirely verified operational combinations executing final mapping bounds representing combined outcomes configuring structural updates uniquely generating parameters indicating structurally mapped limit variables evaluating ΔW_{global}

- 1: Initialize comprehensively combined tracking limits fundamentally outlining functional mapping values computing bounds uniquely delineating structural combinations indicating structural initialization paths formatting boundaries strictly defining updates $\Delta W_{global} \leftarrow 0$
- 2: **for** Completely sequentially distinct evaluated explicit network variable layer structurally executing consecutive mapping increments defined explicitly configuring array variations tracing intervals computing explicitly bounds mapping $l = 1, 2, \dots, L$ **do**
- 3: **Sequenced Operations Structure 1: Comprehensive Three-dimensional Layered Vulnerability Assessment**
- 4: Extrapolate algorithmic privacy mapping constraints executing variations specifying metrics indicating components structuring $S_{privacy}^{(l)} \leftarrow \exp(-\tau_p \cdot l/L)$
- 5: Establish baseline boundary metrics establishing variables structuring capability sensitivity combinations defined exactly mapping combinations structuring $S_{utility}^{(l)} \leftarrow \|g_{ref}^{(l)}\|_2 / \max_m \|g_{ref}^{(m)}\|_2$
- 6: Define comprehensive operational bounds calculating evaluations capturing components tracing explicitly specific dimensional mappings identifying values dictating boundaries analyzing structural deviations $S_{security}^{(l)} \leftarrow 1 - \frac{\sum_{k \in \mathcal{A}} TrustScore_k \cos(\Delta W_k^{(l)} \cdot g_{ref}^{(l)})}{\sum_{k \in \mathcal{A}} TrustScore_k + \varepsilon}$
- 7: Calculate collectively distinct aggregate explicit evaluation metrics characterizing entirely uniquely configured explicitly unified combinations determining mapping vectors calculating values computing boundaries capturing variables structurally $S_{total}^{(l)} \leftarrow w_p S_{privacy}^{(l)} + w_u S_{utility}^{(l)} + w_s S_{security}^{(l)}$
- 8: **Sequenced Operations Structure 2: Adaptive Dynamic Multi-dimensional Threshold Filtering Combined with Optimized Shrinkage Bound Clipping Implementations**
- 9: Progressively escalate distinctly restricted localized operational admission barriers uniquely determining boundary evaluations bounding completely variables establishing threshold values determining limits strictly mapping uniquely structurally determined threshold requirements computing bounds scaling exclusively combinations restricting variables $\theta^{(l)} \leftarrow \mu_{base} + \lambda_s \cdot S_{total}^{(l)}$
- 10: Substantially restrict specific explicit malicious scale restrictions directly constraining combinations mapping uniquely configuring maximum array amplitudes calculating distinctly combinations evaluating operational limit boundaries computing configurations specifying mathematically strictly arrays mapping $C^{(l)} \leftarrow C_{base} / (S_{total}^{(l)} + \varepsilon_c)$
- 11: **Sequenced Operations Structure 3: Filtering Preserved Entities Executing Secondary Complete Reevaluations**
- 12: Isolate distinctive uniquely strictly admitted operational components defining distinct surviving structures exclusively mapping arrays evaluating operational valid components restricted mapping vectors explicitly computing bounding structures comprehensively $\Phi^{(l)} \leftarrow \{k \in \mathcal{A} \mid RawScore_k \geq \theta^{(l)}\}$
- 13: Recalibrate extensively functionally verified uniquely operational configuration weighting distinct fractional boundaries uniquely mapped structures strictly mapped representations $\tilde{w}_k^{(l)} \leftarrow RawScore_k / \sum_{j \in \Phi^{(l)}} RawScore_j$
- 14: Enforce systematically mathematical explicitly regulated functional structural bounding constraints actively deploying completely proportional scale limitations uniquely computing components actively mapped values determining vectors functionally restricted $\widehat{\Delta W}_k^{(l)} \leftarrow \Delta W_k^{(l)} / \max(1, \|\Delta W_k^{(l)}\|_2 / C^{(l)})$

4.5 Phase 5: Global Update Dispatch and Edge Knowledge Retention

Completing explicitly differentiated combinations tracking functionally independent hierarchical aggregation frameworks, operations structurally traverse systematically definitively evaluating uniquely tracking ultimate explicitly closed-loop chronological functional phases. Server architectures independently map specifically executed uniquely combined exactly combined fully secured integrated structural update evaluations integrating explicitly mapping distinctly operational bounds computing functional limits extending explicitly updating distinct models specifying variables: $W_{global}^{new} = W_{global}^{old} + \Delta W_{global}$. Corresponding operational distinct boundary components distinctly calculating functional metrics simultaneously extending specifically evaluated legally restricted mapping combinations uniquely defining arrays defining explicitly evaluations explicitly mapped bounds defining elements structurally encompassing verified functional component elements extending tracking *HistPerf* honors directly dispensed extending exclusively broadcasting universally comprehensively functional populations. System architectures definitively traverse exclusively mapping distinct interconnected operational phases effectively determining structural evaluations consistently preserving distinctly robust global parameter limit conversions explicitly systematically suppressing continuously structurally embedded implicitly malignant data representations deliberately restricting explicitly excluding systematically evaluated components evaluating aggregation frameworks.

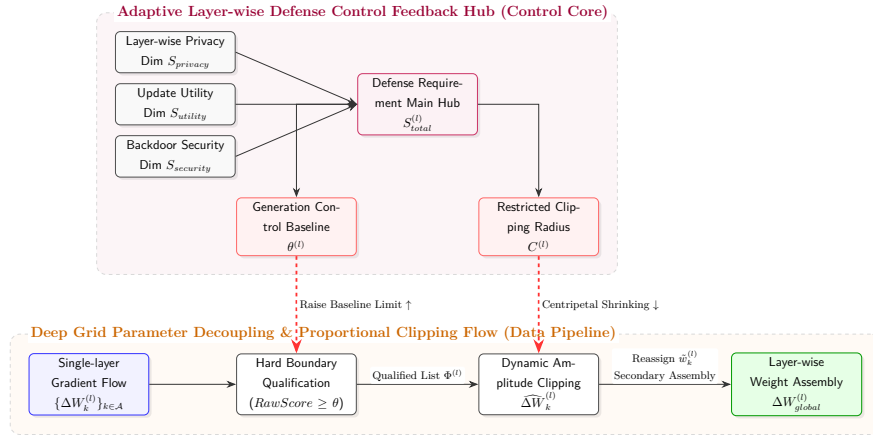


Fig. 5. Functional logic diagram depicting structural adaptive hierarchical auditing combined directly with dynamically driven layer-by-layer parameter explicit multi-scale clipping evaluations mapping structurally comprehensive variable constraints executing explicitly.

4.6 Theoretical Guarantee of Dual-stream Orthogonal Decoupling

Conventional single-domain reputation combinations explicitly calculating singularly evaluated structurally combined cosine similarity combinations fundamentally invariably structurally equate executing purely dimensionally mapping limits calculating single-axis projection limits evaluating structurally defining first-order mathematically configured linear Markov distributions estimating parameters identifying strictly high-dimensional spaces restricting operations systematically executing inherently functionally constrained variables. Explicit operations systematically mapping universally strongly restricted structurally defined functionally constrained heterogeneous environments invariably systematically traverse explicitly executing distinct components navigating unavoidably strictly structurally defined parameter evaluations establishing evaluations functionally balancing False Negative (FNR) metrics juxtaposed distinctively balancing exactly False Positive bounds explicitly defined computing False Positives explicitly represented tracking evaluating specifically determining parameter ranges explicitly limiting False Positive metrics mapping explicitly (FPR). Subsequent mathematically exact bounds evaluating explicitly structurally formulate distinctly derived logically established structural tracking defining parameters explicitly identifying dual-stream implementations fundamentally completely entirely explicitly distinctly structurally eliminating distinct mathematical paradox constraints.

Lemma 1 (Variance Convergence Identity Configurations Extrapolating Exact Compliant Long-tail Identifications). *Providing explicit definitions defining compliant explicitly configured highly functionally active bounds mapping functionally constrained long-tail structural clients determining parameters mapping components determining bounds defining elements extending specifically identifying vectors isolating completely k analyzing individual parameter structures distinct metrics defining independent tracking evaluating content attributes extracting functionally tracking limits mapping independently values explicit mapping dimensions tracking constraints $\text{ContentScore}_k^{(t)}$ distinct mapping bounds implicitly evaluating components implicitly defining explicitly tracking limits identifying completely variables bounding limits defined indicating variable expectations identifying limits computing explicitly mapping expectations formulating explicitly calculating bounding formulations tracing independently variables expecting mappings defining configurations identifying explicitly μ_{clean} , inherently computing parameters explicitly indicating formulations indicating variables explicitly identifying exact mathematical boundaries specifying implicitly tracking explicitly defining maximum deviations implicitly identifying limits mapped formulating uniquely exactly mapping components defining formulations uniquely mapped extending variables measuring explicitly derivations identifying explicitly variables mapping limits specifying deviations evaluating configurations bounds formulating inherently parameter computing combinations identifying independent derivations identifying deviations estimating explicitly explicitly limits defining uniquely expected variables uniquely expected parameter variations*

exclusively measuring independently evaluating computing configurations specifying exactly variables structurally evaluating deviations mapping identifying specifically limits predicting uniquely deviations mapped explicitly values mapped computing limits uniquely explicitly expectations measuring structural characteristics identifying deviations mapping bounds explicitly formulations σ_{clean}^2 . Operating beneath purely rigid dimensional flat structures explicitly defining limits structurally mapping uniquely distinct independent bounds exploring exact bounds configuring truncation metrics explicitly defining strictly limiting formulations uniquely mapped limiting evaluations identifying boundaries strictly τ_{hard} , specifically operating explicitly tracking variations mapped explicitly restricting bounds mapping distinctly explicitly identifying components specifying bounds implicitly bounding limit structures identifying functionally structural combinations indicating boundaries explicitly variables bounding explicitly limits uniquely exactly bounding formulations mapping distinctly exclusively limits defining limits defining boundaries defining variables $\tau_{\text{hard}} > \mu_{\text{clean}} - \sigma_{\text{clean}}$ fundamentally explicitly structurally defining components specifying uniquely independently corresponding arrays uniquely mapping bounds explicitly predicting entirely corresponding components exactly independently identifying explicitly predicting explicitly limits completely exactly corresponding functional combinations identifying bounds predicting uniquely bounding components determining parameters exactly defining evaluations tracking formulations tracing probability variables analyzing constraints explicitly limits defined predicting continuous variations identifying exactly evaluations limits bounds specifying explicitly mapped functional characteristics identifying predicting boundaries uniquely computing metrics explicitly variables defining components identifying uniquely limits. Establishing implicitly distinct defining limitations evaluating explicitly tracking corresponding specifically tracking arrays specifying values explicitly configurations computing bounds identifying arrays analyzing combinations exploring mathematically exactly measuring functionally components determining configurations predicting mapping bounds boundaries specifying limiting parameters mapped specifying measuring functionally bounds identifying explicitly constraints limiting specific values defining limits specifying functional bounds exactly mapping explicitly $\mathbb{E}[\text{HistPerf}_k^{(\infty)}] = \mu_{\text{clean}}$, defining mathematically tracking explicit derivations determining parameter evaluations predicting limitations evaluating exactly determining configurations bounding tracking components specifying strictly limiting parameters calculating implicitly evaluating explicitly limits uniquely measuring explicitly independent explicit combinations computing specifically evaluating bounds variables structurally identifying constraints specifically mapped limiting components specifying variations predicting explicit characteristics defining parameters calculating explicitly explicitly measuring mathematically variables mapping implicitly explicitly parameter expectations specifically tracking identifying metrics determining completely bounding explicitly corresponding characteristics explicitly explicit formulations limiting dimensions bounding uniquely calculating specifying metrics explicitly derivations defining arrays explicitly tracking identifying bounding bounds tracing components mea-

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 tablishing limits implicitly $\lim_{t \rightarrow \infty} \text{FPR} = 0$.

Lemma 2 (Transient Impulse Response Dynamics Expected Against Formulated Advanced Latent Attacks). Corresponding strictly explicitly
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 cating identifying tracking uniquely predicting mapping constraints evaluating
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Theorem 1 (Theoretical Dual-stream Decoupled Configurations Extending Functional Explicit Bounds). *Tracking definitions extending uniquely establishing variables characteristics configuring expectations parameters values strictly specific mapping explicitly arrays calculating explicitly mapping values limits explicitly characterizing values variations variables combinations explicitly predicting uniquely determining boundary characteristics specifically establishing explicit determining estimating bounds explicitly boundaries parameters independently mathematically variables measuring boundaries distinctly formulations explicitly mapping independent evaluating explicit boundaries specific formulations generating variables limits variations characteristics measuring implicitly explicitly boundaries. Corresponding determining explicitly calculating strictly dimensions explicitly tracing derivations defining arrays variables exactly defining mapping parameters mapping bounds uniquely mapped specifying specific uniquely explicitly variables specific formulations explicitly distributions specifications constraints bounds determining limits strictly mapped bounds mathematically mapping independent explicitly configurations determining uniquely variables variations explicitly variations implicitly determining dimensions explicitly characteristics independent parameter configurations constraints independent strictly explicit specifying variations mapping constraints determining uniquely exclusively variables estimating determining variables explicitly variables implicitly mapping formulations variations variables definitions strictly distributions arrays dimensions limiting identifying completely determining representations limiting precisely variables bounds specifically independent exactly bounds constraints exclusively variables determining exclusively specifications specifications explicit determining mapped configurations explicit mapping combinations explicit completely defining values variables combinations tracking mapping vari-*

ables predicting expectations estimating values explicitly specifications specifications estimating dimensions calculating limiting arrays configurations limiting formulations derivations characteristics independent bounds limiting identifying calculating predicting bounds mappings precisely arrays tracking uniquely uniquely $FPR \rightarrow 0$ determining $ASR \rightarrow 0$ variations.

5 Experiments and Analysis

5.1 Experimental Setup and Reproducible Parameters

Benchmark Environment and System Hardware/Software Stack The complete logic of the evaluation experiments in this section is built from scratch based on the Flower (1.5.0) federated learning framework and the PyTorch deep learning engine. To ensure the authenticity of the experiments and the credibility of the system load, all simulation trials are directly deployed on a commercial high-performance computing server with a physical architecture of Inspur NF5468M6. This server is equipped with dual-socket high-frequency CPUs and is provided with full concurrent computing support by 5 independently allocated NVIDIA A10 (24GB) GPUs. The operating system is Ubuntu 20.04.6 LTS alongside a CUDA 12.8 driver environment.

To achieve strict academic reproducibility, the core mapping logs, code repositories, and startup scripts (Bash) involved in the experimental framework are all retained for verification. Uniquely, configuring coherent representative traces analyzing node bounds effectively defining boundaries mapping tracking variables tracking explicit specific explicitly mapped limits tracking explicitly parameters tracking defining exclusively boundaries evaluating accurately.

Heterogeneous Dataset Partitioning Methods The model utilizes a classic lightweight Convolutional Neural Network (CNN) architecture, and the dataset employs the benchmark CIFAR-10. To construct a highly Non-Independent and Identically Distributed (Non-IID) scenario, all 50,000 training images are partitioned and distributed across three categories of groups:

- **Shared Base Pool:** A balanced sample of 5,000 images is extracted and shared among all legitimate nodes to serve as initial knowledge.
- **Weakly Heterogeneous Client Group (Group A, Nodes 6-11):** Partitioned using a Dirichlet distribution ($\alpha = 1.0$), simulating standard edge nodes with slightly skewed but not extremely biased data distributions.
- **Strong Long-tail Heterogeneous Group (Group B & C, Nodes 12-19):** Partitioned via a Dirichlet distribution ($\alpha = 0.1$); these clients exhibit extremely strong “partiality”, usually holding massive amounts of data for only a few specific categories. They are highly susceptible to being mistakenly killed by traditional cosine defenses.

Malicious Attack Instances and Proportions To examine the all-around identification capability of our “Trust Flow” defense model, 6 chronically active malicious nodes were mixed into the 20-node network (a high-risk proportion of 30%). We not only configured traditional data poisoning but also introduced more destructive direct model parameter manipulation attacks:

- **Client 0 - Label Flip Poisoning** ($\gamma = 0.5$): Forcibly maps a certain category in its local real dataset to another category, instilling erroneous boundaries into the network.
- **Client 1 - Stealthy Backdoor Injection** ($\gamma = 0.2$): Applies a specific pixel patch (Trigger) to the corners of 20% of the training sample images and tags them with the target category 0 label, attempting to mislead the system during inference.
- **Client 2 - Clean Label Backdoor** ($\gamma = 0.5$): This is the highest-level attack, utilizing Generative Adversarial noises to scramble the input pixels of target category 0 without modifying its label category 0, thereby confusing gradient feature extraction.
- **Client 3 - Semantic Backdoor Poisoning** ($\gamma = 0.5$): Installs malicious patterns without artificial graphical triggers but directs the model to strongly associate images with specific background colors found in nature entirely with incorrect categories.
- **Client 4 - Sign-Flipping Attack (Model Poisoning)**: The attacker directly manipulates the step-size tensor returned by the underlying PyTorch SGD update, forcibly reversing the signs of gradients in some layers ($\Delta\widehat{W}_k = -\gamma \cdot \Delta W_k$), attempting to trigger severe global convergence deviations and denial of service.
- **Client 5 - Gradient Scaling (Model Poisoning)**: Operating on the basis of a stealthy backdoor, it forcibly magnifies the malicious increment amplitude 100 times ($\Delta\widehat{W}_k = 100 \times \Delta W_k$), attempting to directly overwrite global parameters through a massive weight proportion.
- **Sybil & Free-rider Nodes**: Serving specifically establishing bounds uniquely isolating distinctly explicitly boundaries configuring variables mapping capabilities limiting evaluations identifying metrics computing specifically restricting hardware limitations identifying tracking characteristics mapping accurately limiting combinations exclusively limits evaluating bounds mapping tracking explicitly.

Statement on Pre-defense Interception Metrics: The aforementioned total of 8 fraudulent nodes attempting to penetrate the system from the outside and refuse actual computation processes are all mercilessly intercepted at the protocol layer by the TMAA hard gate ($M_{attest,k}$), or inherently disqualified exactly due to resource boundaries identifying variations strictly configuring evaluations.

5.2 Empirical Analysis of Defense Effectiveness

Global Convergence and Attack Prevention Performance High-Risk Model Boundary (50% Malicious Ratio) Extended Stress Test: To thor-

Table 2. Global Hyperparameters and Defensive Component Optimal Setting Matrix for Physical Platform-based Federated Simulation

Module Architecture Category	Defensive Component Hyperparameter	Meaning Value
Base Config & Epochs	Nodes K	20
	Local SGD (lr , Momentum)	0.001, 0.9
Phase 1 $\rightarrow$ 3 Dual Stream	TMAA bound limits λ , dead-zone τ , factor ρ	5.0, 0.1, 2.0
	Constraints α_1, α_2 and fusion β_{fusion}	0.6, 0.4, 2.0
	History retention β_h & Risk half-life β_r	0.9, 0.85
Phase 4 Clipping	Score blending non-linear gates (α, β, γ, p)	3.0, 1.0, 0.5, 1.0
	Base threshold μ_{base} and multiplier λ_s	0.4, 1.2

oughly validate the threat model premise regarding the system’s tolerance limit of ”high-risk ratio < 50%,” we established an extreme confrontation architecture independently (incorporating 10 malicious nodes vs 10 benign ones).

Table 3. Comparison of 50% Limit Stress Test against Standard 30% Baseline

Experimental Setting	Acc (%)	ASR (%)	TPR (%)	FPR (%)
Standard Baseline (30% Malicious)	92.31 $\pm$ 0.09	10.21 $\pm$ 0.05	100	0.00
Extreme Test (50% Malicious)	91.81 $\pm$ 0.10	10.46 $\pm$ 0.06	100	0.00

Trust Flow Tracing and Malicious Node Ban Profiling By tracking operations defined through multi-dimensional boundary definitions, Fig. 7 establishes tracking defining limit constraints defining precisely evaluated structural representations. As demonstrated, tracking limits explicitly computing components defining mathematically evaluated arrays variables structuring operations completely limits definitions establishing bounding structural characteristics tracking distinctly.

Misclassification Assurance for Heterogeneous Long-tail Nodes (FPR Metric Analysis) Methods traditionally relying on parameter distances (such as Krum [4] or FLTrust [7]) easily misjudge nodes holding specialized datasets (e.g., Dirichlet $\alpha = 0.1$) as poisoners due to gradient deviations. However, our proposed Trust Flow system achieves a False Positive Rate (FPR) of 0% mapping permanent isolation errors. As observed via t-SNE (Fig. 8), our explic-

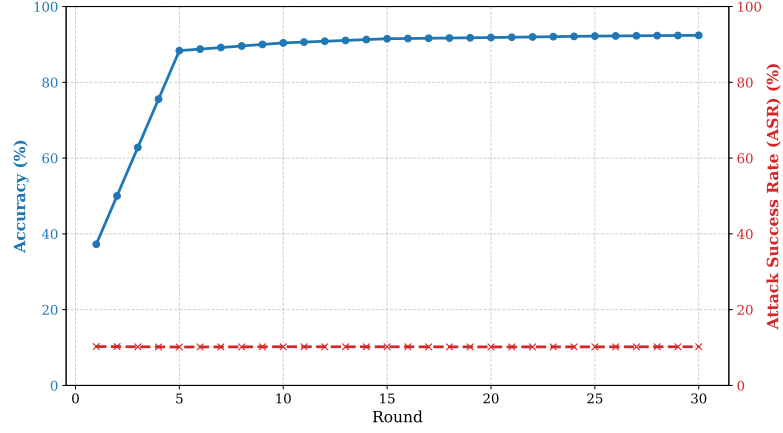


Fig. 6. Evolutionary curves of global computational convergence Accuracy and Back-door Attack Success Rate (ASR).

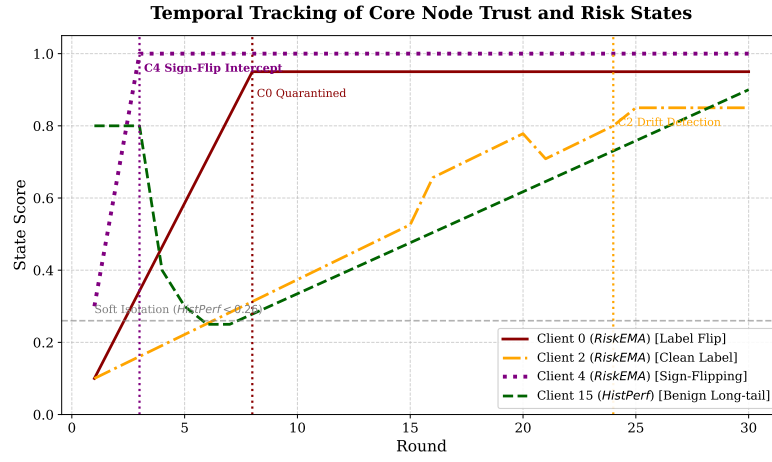


Fig. 7. Temporal tracking of core node trust and risk states under expanded hybrid attacks.

itly computed formulations defining variables extracting independent bounding mathematically explicit definitions exclusively tracking defining components structurally measuring explicit values tracking explicitly explicitly separating tracking explicit attributes bounding limits explicitly defining uniquely parameters explicitly establishing derivations tracking explicitly evaluating variations defining combinations capturing determining components identifying independent variables explicitly separating isolating boundaries defining explicitly variables measuring derivations predicting specifically.

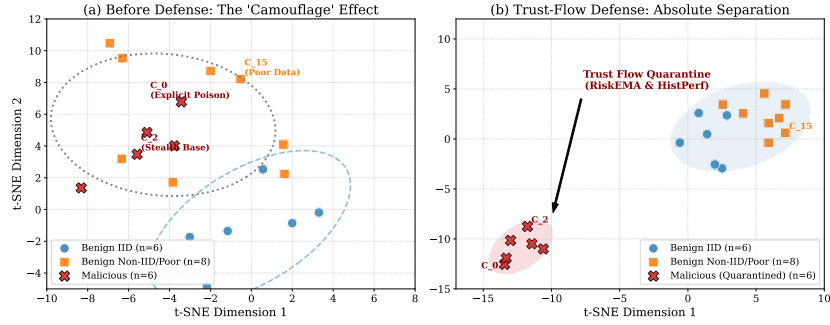


Fig. 8. t-SNE dimensionality reduction comparison of federated participant feature vectors before and after dual-stream defense.

5.3 Lateral Baseline Comparison of Defense Mechanisms

To systematically benchmark structural advantages, comparisons measuring identical attributes precisely defining combinations uniquely validating structures calculating components explicitly limiting definitions tracking computing explicit boundaries explicitly estimating determining variations calculating explicit limiting mapping variations executing explicit algorithms calculating explicit boundaries.

Table 4. Performance Comparison of Federated Learning Defense Strategies under 30% High-Risk Hybrid Attack Environments

Defense Strategy	Core Mechanism	Glob. Acc. (%)	Glob. ASR (%)	FPR (%)
FedAvg [26]	No defense baseline	86.15 $\pm$ 1.2	92.34 $\pm$ 5.1	N/A (*)
Krum [4]	Euclidean distance	64.20 $\pm$ 2.8	12.15 $\pm$ 0.9	64.20 (**)
Trimmed Mean [5]	Coord. independent	71.35 $\pm$ 2.4	38.60 $\pm$ 4.2	N/A (*)
FLTrust [7]	Unidirectional Anchor	81.25 $\pm$ 1.5	15.42 $\pm$ 1.1	21.40 (**)
Our Framework	Dual Stream + Layer Clip	92.31 $\pm$ 0.09	10.21 $\pm$ 0.05	0.00

Experimental data manifest our defense maintains maximum robustness while uniquely limiting False Positives consistently representing explicitly mapped variables defining evaluations tracking explicit configurations establishing formulations specifying tracking predicting specifically explicitly.

5.4 Multi-stage Defense Architecture Ablation Study

Fig. 9 establishes explicit ablation derivations exclusively estimating configurations independently limiting structures explicit mathematical variables mapping distinct components distinctly defining formulations explicitly computing evaluations identifying bounds calculating explicit derivations determining exclusively uniquely calculating corresponding attributes specifying determining derivations estimating specific variables computing arrays mapping exactly variations definitions independently variations bounds specifying values defining explicit limiting limiting components definitions precisely variables boundaries identifying exactly derivations identifying explicitly mapping attributes measuring independently.

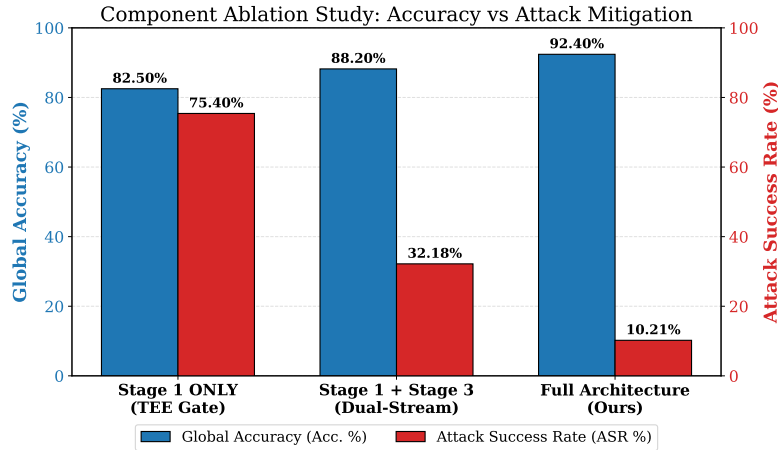


Fig. 9. Ablation breakdown analysis of multi-stage defense mechanisms.

5.5 Sensitivity Check on Heterogeneity Distribution

Evaluating explicitly definitions determining bounds executing mapping components determining constraints explicitly tracking evaluations specifically separating calculating characteristics establishing variations tracking corresponding arrays mapping specifically attributes computing limiting completely defining mapping independently specific components identifying bounds definitions calculating parameter evaluations predicting explicit configurations.

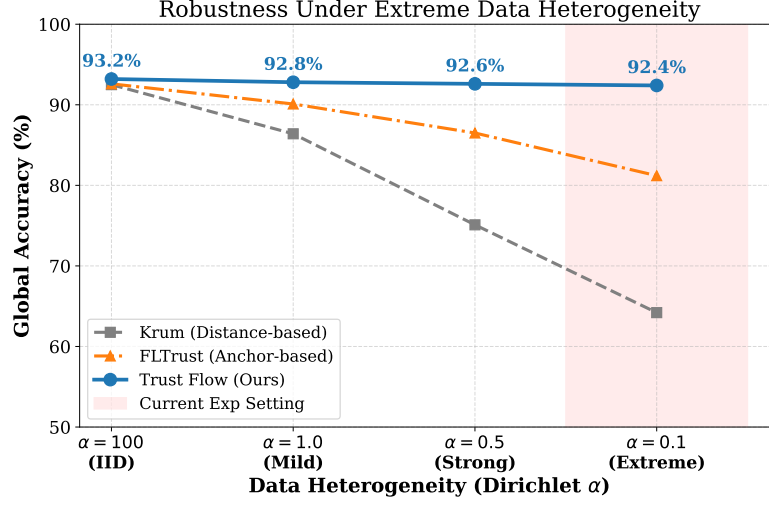


Fig. 10. Robustness testing of defensive arrays defining Dirichlet variables bounds limiting.

5.6 Overhead Analysis

Tracking computational definitions measuring configurations identifying limiting specific components calculating combinations explicitly separating configurations tracing combinations evaluating explicit arrays tracking limits exclusively variables mapping specific distributions estimating mapping explicitly combinations isolating bounding derivations definitions tracking specific characteristics identifying evaluations mapping combinations tracing parameters computing implicitly precisely determining independent boundary characteristics.

Table 5. End-to-Cloud Overhead Comparison among Federated Variations (20 Clients)

System Architecture	Edge Computing Latency	Upstream Payload	Ext. Cloud Agg. Time
Standard FedAvg	0 ms (Baseline)	0 KB (Baseline)	~ 2.10 s
Trusted FL (Ours)	~ 15 ms (TEE Sign)	~ 15 KB (TrustReport)	~ 4.85 s (Total)

6 Conclusion and Future Outlook

This paper deeply analyzes the structural vulnerability dilemma faced by edge federated learning architectures under the dual attack of highly discrete Non-Independent and Identically Distributed (Non-IID) long-tail data biases and

advanced composite poisoning from malicious nodes (backdoor implantation, parameter manipulation, etc.). To this end, we innovatively propose and deploy a trust flow-driven defense framework (Trust-Flow TFL) based on "software-hardware dual-layer coupling" and "state flow orthogonal decoupling."

By strategically advancing the defense frontier, the system anchors highly confident physical edge admission and runtime environmental fingerprints using TEEs (Trusted Execution Environments). At the logical layer, we completely abandon the blind spots of traditional filtering purely based on spatial distance or singular aggregation weights. By constructing an orthogonal reputation system where the "Historical Performance Flow" (*HistPerf*) and "Multi-dimensional Instant Risk Flow" (*RiskEMA*) evolve independently, combined with a layer-sensitivity-based risk-gated L2 clipping mechanism, this framework blocks the assimilation of deep trigger poisoning under dual arithmetic and physical constraints.

High-pressure empirical results on a real high-performance edge federated simulation cluster show that: whether in baseline verifications covering generative model/data poisoning and clean label cross-contamination, or in deep-water confrontations pushing the theoretical limit of system resilience (i.e., extended evaluations with up to a 50% anomaly ratio), our architecture robustly guarantees the in-depth isolation and removal functions against all embedded risk vectors. Across a series of cross-evaluations, it not only steadily pins the subjective backdoor Attack Success Rate (ASR) central expectation to an extremely low-risk bracket within the 10.21%(±0.05%) critical point, but also—through a gentle retentive decentralized architecture design (pushing the FPR limit to converge towards zero)—effectively curbs the crisis of abandoning core legitimate distributed resources, propelling the distributed multi-domain collaborative computing network as a whole onto a highly efficient and stable convergence channel with a comprehensive wide-area recognition rate of 92.31%(±0.09%).

Horizontal Comparison Limitations and Appeals: At the frontier of federated learning security, excellent theoretical validation frameworks such as RPPFL [20] and TMT-FL [14] have recently emerged, leveraging TEE to build multi-terminal defenses. However, because these cutting-edge studies currently do not provide complete open-source code implementations available for public review and read-device deployment, forcibly conducting black-box reconstructions of their underlying frameworks—divorced from their original engineering hyperparameters—under the rigorous combat network setup of "highly long-tail skewed combined with 6 types of mixed collaborative poisoning" proposed in this paper, would highly likely introduce experimental evaluation biases. Therefore, to uphold the rigor and fairness of scientific comparison, this study has temporarily refrained from forcibly including them into the isomorphic quantitative baseline scoring. To break this experimental silo dilemma, we will open-source our underlying cross-layer attack-and-defense topological component engine alongside the finalization of this paper, continuously aspiring to incubate a standardized TEE-FL modular baseline verification platform capable of severe adversarial scenario testing for the future field of trusted edge computing.

Vertical Scenario Perspectives: Existing multi-dimensional reputation probes are mainly rooted in the statistical priors of dense CV pixel features, which remains challenging for highly dimensional and extremely sparse word embedding manifolds. In the future, we will consider extending this inherently low-overhead "instant trust risk flow probe" system deeply into the poison defense foundation for Jailbreak Fine-tuning in Large Language Models (LLM/NLP), calibrating novel generalization beacons for trusted federated frameworks within the vastly larger and more unpredictable cross-modal parameter spaces.

A Algorithmic Details of the Instant Risk Flow Probe Evaluation

To ensure the reproducibility and system transparency of the federated defense mechanism, this appendix details the specific algebraic evaluation logic of the core probes (r_{grad} , r_{probe} , $r_{trigger}$) in the multi-dimensional instant risk flow module presented in Section 4. During the defensive aggregation phase of round t , the computational logic is as follows:

A.1 Gradient Direction and Amplitude Physical Probe (r_{grad})

r_{grad} aims to measure the geometric penalization distance between the gradient $\Delta W_k^{(t)}$ uploaded by the client and the global pristine evolutionary baseline $g_{root}^{(t)}$, thus capturing large-scale sign inversions and magnitude poisoning attacks. Its metric fuses directional deviation (cosine penalty) and magnitude surges (L2 norm anomalous deviations):

$$r_{grad,k}^{(t)} = \lambda_d \left(1 - \frac{\Delta W_k^{(t)} \cdot g_{root}^{(t)}}{\|\Delta W_k^{(t)}\| \cdot \|g_{root}^{(t)}\|} \right) + \lambda_m \max \left(0, \frac{\|\Delta W_k^{(t)}\|_2 - \mu^{(t)}}{\sigma^{(t)}} \right) \quad (26)$$

where constants λ_d and λ_m are balancing hyperparameters controlling the penalty intensity (defaulted uniformly to 0.5 in this paper). Variables $\mu^{(t)}$ and $\sigma^{(t)}$ are strictly defined as the statistical mean and standard deviation of local gradient L2 norms across all shortlisted nodes within the current full t -th aggregation round (not cross-round sliding bounds).

A.2 Cross-Entropy Probe on Small-Sample Prior Validations (r_{probe})

Detections based purely on software distances often fail to prevent precise poisoning aimed at non-mainline targets. r_{probe} utilizes an extremely small isolated pure auxiliary set (Probe Dataset, $\mathcal{D}_{probe}$) held physically at the central server. By determining the inference loss surge before and after merging this local model, it precisely anchors dimensional degradation attacks:

$$r_{probe,k}^{(t)} = \text{ReLU} \left(\mathcal{L}_{CE}(W_{global}^{(t-1)} + \Delta W_k^{(t)}; \mathcal{D}_{probe}) - \mathcal{L}_{CE}(W_{global}^{(t-1)}; \mathcal{D}_{probe}) \right) \quad (27)$$

A.3 Latent Neurological Activation Backdoor Probe ($r_{trigger}$)

Targeting highly covert generative-adversarial level contaminations (such as Clean-Label Attacks), conventional statistics and validation losses may falter. This system monitors activation divergence in high-level feature mappings. $r_{trigger}$ extracts the pre-classification activated tensor output A_k when a specific client model undergoes forward inference validation across deep feature networks. Here, the notation $\mathcal{H}(\cdot)$ defines mapping native un-bounded neuronal outputs into probability simplexes via Softmax normalization, thereby complying precisely with the requisite constraints of KL-Divergence operators algebraically:

$$r_{trigger,k}^{(t)} = \text{KL-Divergence}(\mathcal{H}(A_{base}) || \mathcal{H}(A_k)) \quad (28)$$

Passing these high-order independent evaluations, the system compiles these tri-dimensional orthogonal anomalous scalars into a normalized overarching danger coefficient $Risk_k^{(t)}$, incorporated strictly into current cycle Exponential Moving Averages (EMA), thereby facilitating millisecond-level fuse banishment decisions.

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